MAKRAND GHODKE

M.Sc. student in Information & Communication Engineering | AI, machine learning and software

Darmstadt, Germany | Open to working student and HiWi roles in ML, AI and data | Up to 20 hours a week during the semester

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SUMMARY

Master's student at TU Darmstadt, expected to graduate in autumn 2027, working on machine learning and software engineering. I build and test software around ML models, mostly in Python and TypeScript. In a course project I found why an image classifier was stuck at 30% accuracy and fixed it, which took it to 100%. I later deployed that backend with Docker and added automated tests.

PROJECTS

DigiGuide, landmark recognition app (course project) *Oct 2025 - Feb 2026*

- Co-developed the image classification API: CLIP embeddings and FAISS similarity search behind a FastAPI endpoint, serving live camera input for 5 landmark classes.
- Led data collection and image preprocessing. Traced the 30% accuracy to a missing BGR-to-RGB conversion; after the fix, all 5 test classes reached 100%.

DigiGuide backend: deployment and tests github.com/MakrandGhodke/digiguide *Oct 2026*

- Packaged the backend in Docker and deployed it on Render. The first deploy ran out of memory, so I moved the model download into the Docker build and removed the unused text encoder.
- Wrote 38 pytest tests with the model and an external API mocked, and set up GitHub Actions to run them on every push.
- The tests found three bugs: a path-traversal hole, a 500 error on invalid uploads and a hard-coded JWT secret. I fixed all three, each with its own test.

SpendWise, household finance app (PWA) github.com/MakrandGhodke/SpendWise *Mar 2026 - May 2026*

- Built on my own with Next.js, TypeScript and Supabase (PostgreSQL). Row Level Security keeps each user's data separate.
- Imports bank CSV files (Revolut, N26, Sparkasse, Deutsche Bank and others) with automatic format detection, category rules and hash-based duplicate detection. Balance updates run in SQL stored procedures, so balances stay consistent.
- Works offline: changes are saved locally and synced when the connection returns. Installable as a PWA, built as an iOS app with Capacitor, deployed on Vercel, with a 6-month trend dashboard.

EXPERIENCE

Web Developer Intern, Exposys Data Labs, Pune *Jul 2022 - Aug 2022*

- Built and tested responsive web pages (HTML, CSS, JavaScript) in Chrome, Firefox, Safari and Edge, in an Agile team of five.

Operations Support Technician, Flextronics Technologies, Pune *Jul 2021 - Dec 2021*

- Redesigned the Excel tracking tools used for production data reporting. Manual entry errors went down and teams could see the same data.

EDUCATION

M.Sc. Information and Communication Engineering *Oct 2024 - expected autumn 2027*

Technische Universität Darmstadt. Focus: AI and machine learning. Courses: Introduction to AI, Data Science 1 and 2, Explainable and Ethical AI.

B.Tech Electronics and Telecommunication *Nov 2020 - Jul 2023*

Bharati Vidyapeeth Deemed University, Pune. CGPA: 9.06 / 10

SKILLS

Machine learning: Python, PyTorch, CLIP, FAISS, scikit-learn, pandas, NumPy, computer vision, model evaluation

Backend and delivery: FastAPI, REST APIs, Docker (familiar), GitHub Actions (CI), pytest, Git, GitHub, Render, Vercel

Web and data: TypeScript, JavaScript, Next.js, React, Node.js, Flutter/Dart, SQL, PostgreSQL (Supabase), Tailwind CSS

Other: Java, C++, Jupyter Notebook, VS Code, Agile/Scrum

LANGUAGES

English: C1 (fluent) | German: A2-B1 (basic) | Hindi: native | Marathi: native